

BAENGYEONGDO, Korea, Republic of

WMO: 471020

Lat: 37.974N

Lon: 124.713E

Elev: 125

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	19.2	2.0	6.1	21.2	4.4	6.9	22.1	45.2	28.7	35.0	30.1	12.2	290	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	82.9	76.1	80.7	74.6	78.8	73.4	78.1	80.8	76.7	79.3	75.4	77.8	7.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.8	79.8	75.9	136.2	78.5	74.5	129.7	77.3	41.8	81.0	40.4	79.4	39.1	77.9	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.7	26.5	22.5	12.0	86.8	5.1	2.2	8.3	88.3	5.3	89.6	2.4	90.9	-1.3	92.5
			10.3	79.2	4.6	2.3	7.0	80.9	4.3	82.3	1.7	83.6	-1.6	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	29.6	32.1	39.0	48.6	58.5	66.5	72.5	75.1	68.7	58.9	46.1	34.2
	DBStd	16.55	6.23	6.25	5.24	5.22	4.94	3.95	3.61	3.75	3.55	5.15	6.85	6.88
	HDD50	2214	631	501	342	87	1	0	0	0	0	7	154	491
	HDD65	5286	1096	921	806	494	209	30	1	0	11	195	568	955
	CDD50	3157	0	0	1	44	264	496	696	777	560	282	37	0
	CDD65	753	0	0	0	0	7	76	232	312	121	5	0	0
	CDH74	2524	0	0	0	0	7	87	700	1601	128	1	0	0
	CDH80	256	0	0	0	0	0	1	58	196	1	0	0	0
Wind	WSAvg	10.6	10.8	10.8	11.8	12.9	12.1	9.8	9.6	8.6	7.9	9.4	11.5	11.6
Precipitation	PrecAvg	32.2	0.5	0.7	0.7	2.0	2.8	3.0	8.6	6.5	3.4	1.4	1.6	0.9
	PrecMax	49.8	1.1	2.6	2.8	3.9	5.4	8.0	15.7	12.9	14.3	3.7	6.1	1.7
	PrecMin	16.0	0.1	0.0	0.1	0.2	0.1	0.6	2.9	1.4	0.2	0.0	0.4	0.3
	PrecStd	7.9	0.3	0.6	0.7	1.1	1.7	1.8	3.7	3.4	4.1	1.1	1.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	48.5	55.2	66.0	74.9	78.9	84.7	86.3	79.6	72.0	62.2	51.3
		MCWB	40.7	42.8	43.5	51.2	57.4	65.2	76.9	78.5	69.6	63.5	56.9	48.1
	2%	DB	42.3	44.3	50.7	62.0	71.3	76.5	81.4	84.1	77.1	69.1	59.6	48.5
		MCWB	38.4	40.0	43.9	49.8	55.9	65.0	75.1	77.1	69.3	62.1	54.9	44.9
	5%	DB	40.1	42.0	48.0	59.0	68.8	74.4	79.2	82.2	75.4	66.9	57.4	45.8
		MCWB	35.9	38.0	42.1	48.3	56.2	64.9	74.1	75.8	68.0	60.7	53.3	41.3
	10%	DB	38.1	39.9	45.8	56.4	66.4	72.5	77.5	80.3	73.6	65.1	55.4	43.7
		MCWB	34.0	36.2	40.8	47.1	55.3	64.7	73.5	74.9	67.0	59.3	51.1	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	45.4	48.5	56.4	62.9	72.0	79.2	80.5	74.4	66.6	59.3	49.3
		MCDB	43.7	47.2	50.4	61.4	68.3	75.0	80.9	84.2	76.5	69.2	61.0	50.7
	2%	WB	39.1	40.9	45.9	53.0	61.0	69.6	77.2	78.9	71.8	64.4	56.6	45.5
		MCDB	41.7	43.2	48.8	57.6	66.2	72.3	79.4	81.9	74.2	67.4	58.4	48.2
	5%	WB	36.7	38.8	43.6	51.1	59.5	68.2	76.0	77.7	70.3	62.2	53.9	42.2
		MCDB	39.5	41.3	46.4	55.2	64.5	71.3	78.1	80.5	73.0	65.4	56.4	45.4
	10%	WB	34.2	36.7	41.7	49.3	58.0	67.0	74.7	76.5	68.9	60.6	51.8	39.6
		MCDB	37.8	39.4	44.8	53.7	63.1	70.2	76.7	79.2	72.3	64.1	55.4	43.3
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	8.5	9.7	10.7	9.2	7.4	7.5	7.9	7.9	7.4	7.3
		MCDBR	8.6	9.5	11.7	13.2	13.8	11.8	9.7	9.2	9.6	9.3	7.7	8.8
		MCWBR	8.4	8.2	8.1	7.9	7.6	6.7	5.5	5.4	6.3	6.8	7.5	8.8
	5% WB	MCDBR	8.6	9.2	10.6	11.2	11.6	8.7	7.6	8.0	7.8	8.9	7.5	9.1
		MCWBR	8.7	9.0	9.1	8.5	7.6	5.8	5.5	5.3	6.6	7.9	8.5	9.7
Clear-Sky Solar Irradiance	taub	0.439	0.494	0.554	0.605	0.624	0.706	0.682	0.585	0.515	0.526	0.507	0.446	
	taud	1.944	1.841	1.740	1.682	1.696	1.607	1.706	1.920	2.017	1.866	1.862	1.932	
	Ebn at Noon	218	223	224	222	221	204	207	225	233	211	193	204	
	Edh at Noon	44	55	67	75	76	83	75	59	50	53	48	42	
All-Sky Solar Radiation	RadAvg	702	998	1388	1601	1761	1620	1386	1492	1415	1132	749	609	
	RadStd	55	74	115	110	163	195	187	181	121	78	87	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	+2.59	N/A	N/A	N/A	N/A
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	+86

Nomenclature: See separate page